

Mattia Volpato

Data Engineer | Software Engineer • Algorithms & Data Structures • Distributed Systems

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Summary

Data Engineer with a strong software engineering focus, working on distributed systems and data-intensive applications. Experience developing scalable data platforms and distributed processing workloads using **Python**, **PySpark**, **Databricks** and cloud technologies, alongside software engineering experience in **Java**, AWS and microservices. Strong computer science background in algorithms, data structures and graph theory.

Education

University of Milano Bicocca, Master of Science in Computer Science – Milan, Italy Oct 2023 – Sept 2025
110/110 cum laude

- Thesis: [Delay Prediction in Supply Chains — A Hybrid Graph-Based and Machine Learning Approach](#). Developed at Cefriel as part of the M4ESTRO Horizon Europe project.

University of Milano Bicocca, Bachelor of Science in Computer Science – Milan, Italy Sept 2020 – July 2023
110/110 cum laude

- Thesis: [Wavefront Algorithm for Recombination Detection](#). Implemented a Rust prototype for graph-based sequence alignment.

Experience

Data Engineer, Quantyca | Data at Core – Monza, Italy Oct 2025 – present
11 months

Data Engineer working on scalable data-intensive systems and enterprise data platforms for clients including [Prelios](#) and [TeamSystem](#), with a focus on distributed processing, software development and maintainable data solutions.

- For Prelios, developed a configuration-driven alerting system on **Databricks**, supporting 32 configurable alert logics and 54 deployed alerts. Developed reusable **Python** libraries for Jinja-based templating and Azure Service Bus integration, and a distributed **PySpark** pipeline to render, execute and persist alert results, with typical execution times of 10–20s per alert and publishing 1K–10K messages through Azure Service Bus.
- For TeamSystem, developed and maintained 9 enterprise data products within a Data Mesh, processing gigabytes-to-terabytes of data from sources including Jira, GitHub and security platforms. Used **Databricks**, **PySpark**, **SQL** and Terraform to build and improve data pipelines, datasets, metrics and dashboards, with a focus on performance and resilience.

Software Engineer Intern, Cefriel – Milan, Italy Feb 2025 – Sept 2025
8 months

Software engineering internship within M4ESTRO, a Horizon Europe research project focused on resilient manufacturing ecosystems and intelligent supply chains.

- Designed and evaluated a graph-based model for shipment delivery-time prediction in a single-manufacturer supply chain network, combining graph theory, Markov chains and machine learning.
- Implemented and deployed the model as an working prototype within the M4ESTRO platform using **Python**, AWS Lambda and AWS CDK, contributing across the software lifecycle from requirements analysis through deployment, using real industrial data.

Freelance Software Engineer Consultant, Andreoli Nastri S.n.c. – Cologno Monzese, Italy July 2024 – June 2026
2 years

Developed a complete desktop business application for a small manufacturing SME, working directly with the customer from requirements analysis through architecture, implementation and deployment. Designed the solution around the expected usage, avoiding unnecessary architectural complexity and costs.

- Designed and developed the application using **Java**, **Spring Boot** and Swing, integrating the user interface, business logic and persistence layers, with CI/CD workflows for automated testing and deployment.
- Designed and implemented the relational data layer using Spring Data JPA, Hibernate and PostgreSQL, with Neon Serverless PostgreSQL and Flyway for database deployment and migrations.

Certifications

Databricks Certified Data Engineer Associate Dec 2025

Projects

AeroFlux

Nov 2024 – May 2025

7 months

Cloud-native distributed platform for the safe integration of drones into European airspace, developed as a Master's software engineering project with a focus on distributed systems and safety-critical software.

- Designed and implemented the flight-plan computation module using graph traversal algorithms for real-time, safety-critical route planning.
- Contributed to microservices for flight authorization, geo-awareness and UAV simulation, applying hexagonal architecture to decouple business logic from external integrations.
- Set up CI/CD pipelines to automatically build and deploy services to a Kubernetes cluster, later consolidating to Docker-only when the underlying VM infrastructure was decommissioned.

Delay Prediction in Supply Chains — A Hybrid Graph-Based and Machine Learning Approach

Feb 2025 – Sept 2025

8 months

Master's thesis developed at Cefriel within the Horizon Europe M4ESTRO project, combining graph-based modelling and machine learning for shipment delay prediction using industrial supply chain data.

- Modelled supply chain networks as graphs and developed a hybrid prediction approach combining graph-based methods, Markov chains and machine learning.
- Evaluated the approach against statistical and graph-based methods using industrial data from a manufacturing supply chain.

Templisafe

Mar 2026 – present

6 months

Open-source **Python** library for safe, validated and configurable templating workflows, inspired by Infrastructure-as-Code principles.

- Designed an extensible software architecture with a minimal public API, schema validation, configurable data sources and pluggable template engines.
- Applied modern **Python** engineering practices, including Pydantic, type checking, automated testing, code quality checks, CI/CD pipelines and package publishing workflows.

Wavefront Algorithm for Sequence Alignment

Feb 2023 – July 2023

6 months

Bachelor's thesis implementing a Rust prototype of the Wavefront algorithm for sequence alignment over graph-based genome representations.

- Implemented the Wavefront sequence alignment algorithm in Rust for graph-based genome representations.
- Explored algorithmic optimizations with a focus on computational efficiency and memory usage.

Skills

Programming: Python, Java, C++, Rust, SQL

Distributed Systems & Data Processing: Apache Spark, **Databricks**, Azure Service Bus, ActiveMQ

Software Engineering: Spring Boot, Hibernate, FastAPI, SQLAlchemy, REST APIs, Microservices

Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, Terraform, Linux

Databases: PostgreSQL, MongoDB, BigQuery

Languages

Italian: Native

English: Professional working proficiency (B2)